BACKTEST PLAN AND VERIFIED EXECUTION FLOW

How 192 screening parameter sets are evaluated from January 2022 through fixed return horizons

Verdict: The proposed flow is correct, with one important detail: for each parameter set, screen type, and security, the implementation retains only the earliest qualifying selection that completes all required confirmations.

Verified Flow

1. **Create 192 parameter sets:** Build the Cartesian product of six selectable screening parameters. Daily and weekly moving-average tolerances remain fixed.
2. **Build historical features:** Read production S3 daily price data and fundamental growth histories, then build daily and completed-week feature Parquet files with DuckDB.
3. **Find A-F and A-H selections:** Evaluate every parameter set from January 1, 2022. A-F becomes actionable after the next-day F confirmation; A-H becomes actionable after G and the following completed-week H confirmation.
4. **Keep the earliest selection per security:** Within each parameter set and screen type, later qualifying signals for the same gvkey/iid security are discarded.
5. **Calculate fixed-horizon returns:** Measure each retained selection at 6 months, 1 year, and 2 years using the first available trading price on or after the calendar horizon.
6. **Compare parameter-set performance:** For each screen and horizon, aggregate completed sample count, average return, median return, and win rate; then combine the three average-return ranks.

Parameter Grid

Parameter	Choices	Condition
Annual growth threshold	2%, 3%	A
Quarterly growth threshold	2%, 3%	B
Annual periods	2, 3 years	A
Quarterly periods	2, 3, 4 quarters	B
Volume-ratio threshold	2x, 3x, 4x, 5x	C / D
Volume-surge minimum days	2, 3 days	D
Daily MA tolerance	1% fixed	E / F
Weekly MA tolerance	2% fixed	G / H

Combination count: 2 x 2 x 2 x 3 x 4 x 2 = 192 parameter sets.

Causal Selection Timing

No pre-confirmation return: Entry price and return measurement begin only when the required confirmation is observable. Price movement before the actionable selected date is excluded.

Stage	A-F	A-H
A-E signal	Daily signal date	Daily signal date
F confirmation	Next trading row	Next trading row
G confirmation	Not required	First completed official week on/after F
H confirmation	Not required	Following completed official week
Actionable selected date	F confirmation date	H confirmation date
Entry price	F-confirmation adjusted close fallback	H-confirmation weekly close

Selection Unit

The stored observation is not every signal event. It is the earliest qualifying security selection for:

- one parameter set
- one screen type: A-F or A-H
- one security identified by gvkey and iid

Consequence: A stock that qualifies again later under the same parameter set and screen does not create another outcome. This avoids repeated-event concentration, but it means the study evaluates first-selection performance rather than every trading opportunity.

Fixed-Horizon Return Calculation

Horizon	Price date used	Return formula
6 months	First trading date on/after selected date + 6 calendar months	(price_6m / entry price - 1) x 100
1 year	First trading date on/after selected date + 1 calendar year	(price_1y / entry price - 1) x 100

Horizon	Price date used	Return formula
2 years	First trading date on/after selected date + 2 calendar years	(price_2y / entry price - 1) x 100

If a selection has not yet reached a horizon, its horizon return remains null and is excluded from that horizon's comparison.

Performance Comparison

- A-F and A-H are compared separately because they have different confirmation timing and entry dates.
- Each horizon is ranked independently by average return after applying a minimum completed-sample threshold.
- The final combined score is the mean of the 6-month, 1-year, and 2-year average-return ranks; lower is better.
- Combined eligibility requires sufficient completed observations at all three horizons.

Latest Verified Run

Screen	Selections	Unique securities	6m complete	1y complete	2y complete
A-F	3,698	92	3,364	3,050	2,436
A-H	1,928	47	1,748	1,576	1,292

All 192 parameter sets produced stored outcomes for both screens. Validation reported zero missing core price outcomes, zero confirmation-timing errors, zero inconsistent fixed-horizon rows, and zero invalid horizon dates.

What This Backtest Answers

- Which parameter sets historically selected securities with stronger first-selection returns at each fixed horizon?
- Which parameter sets performed consistently across 6-month, 1-year, and 2-year comparisons?
- How much completed sample coverage supports each ranking?

What It Does Not Yet Prove

- A higher rank does not prove statistically significant superiority because parameter sets often reuse the same securities and dates.
- The current design does not simulate repeated entries, portfolio sizing, transaction costs, or overlapping-position capital constraints.
- Fundamentals use datadate <= signal date because filing/publication availability dates are not present; this reduces but does not fully remove look-ahead risk.
- Fixed-horizon outcomes describe hypothetical holding periods, not an implemented exit strategy.